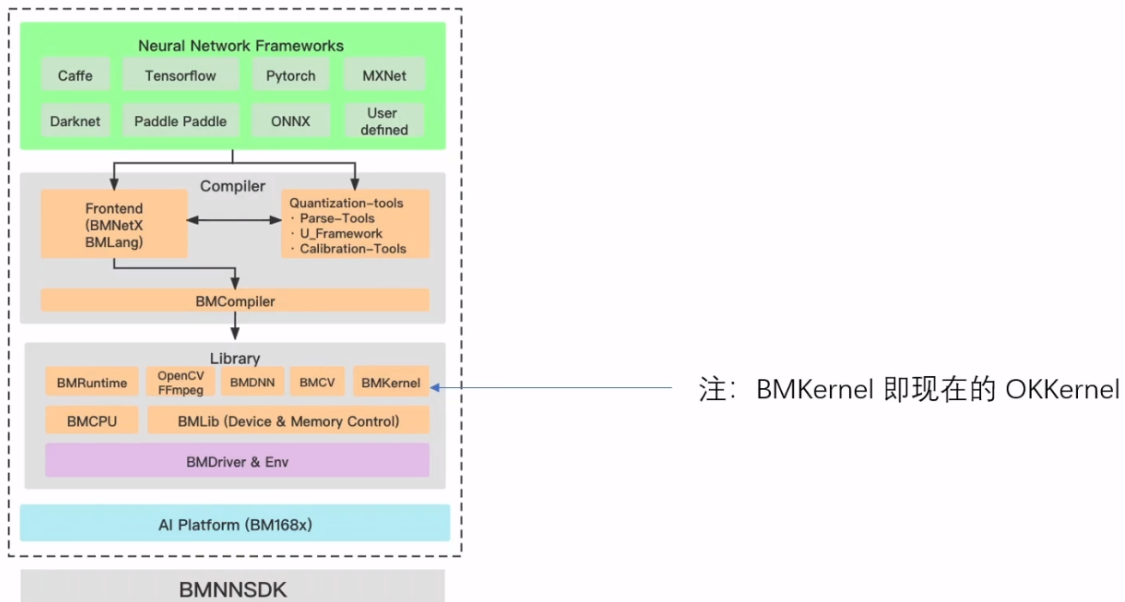


okkernel

要求

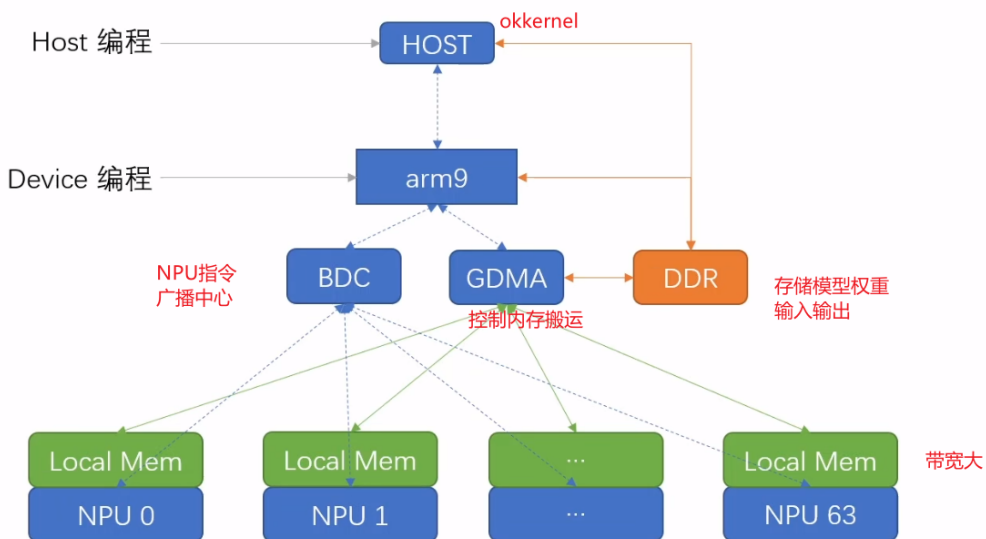
- 芯片架构——计算单元的并行度、层次存储结构
- 熟悉硬件指令集



相关硬件

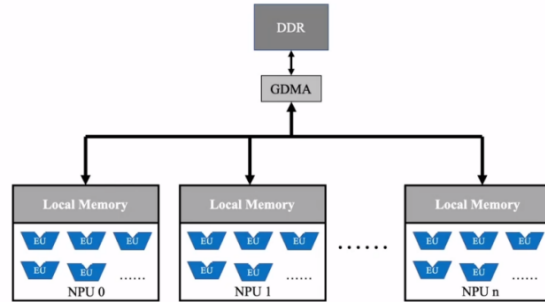
okkernel: 提供底层原子接口，封装通讯机制

全局DDR核本地内存的关系 (bm1684) 单指令 多数据,



众核架构

每个核称作一个神经网络处理单元(NPU)，每个NPU都有独立的本地存储(Local Memory)和多个计算单元(EU)。所有NPU以单指令多数据(SIMD)方式完成一个计算操作，只有在NPU本地存储中的数据才能被EU访问



示例

两个copy

内存接口

okk_gdma_32bit_cpy_S2LDDR → LocalMem

```
void okk_gdma_32bit_cpy_S2L(local_addr_tdst_addr, system_addr_tsrc_addr, const dim4 *shape, const dim4 *dst_stride, const dim4 *src_stride)
```

Copy tensor from system memory to local memory for 32-bit data type.

$$dst(n, c, h, w) = src(n, c, h, w)$$

- Parameters:
- `dst_addr` - Address of the destination tensor in local memory.
 - `src_addr` - Address of the source tensor in system memory.
 - `shape` - Pointer to the shape of the destination and source tensors.
 - `dst_stride` - Pointer to the stride of the destination tensor.
 - `src_stride` - Pointer to the stride of the source tensor.

Remarks

- The data type of the destination and source tensors is 32-bit.
- If `dst_stride` is NULL, the destination tensor is in the 128-Byte Aligned Layout.
- If `src_stride` is NULL, the source tensor is in the Continuous Layout.
- `dst_stride->w` and `src_stride->w` are only allowed to be one.

okk_gdma_32bit_cpy_L2S

```
void okk_gdma_32bit_cpy_L2S(system_addr_tdst_addr, local_addr_tsrc_addr, const dim4 *shape, const dim4 *dst_stride, const dim4 *src_stride)
```

Copy tensor from local memory to system memory for 32-bit data type.

$$dst(n, c, h, w) = src(n, c, h, w)$$

- Parameters:
- `dst_addr` - Address of the destination tensor in system memory.
 - `src_addr` - Address of the source tensor in local memory.
 - `shape` - Pointer to the shape of the destination and source tensors.
 - `dst_stride` - Pointer to the stride of the destination tensor.
 - `src_stride` - Pointer to the stride of the source tensor.

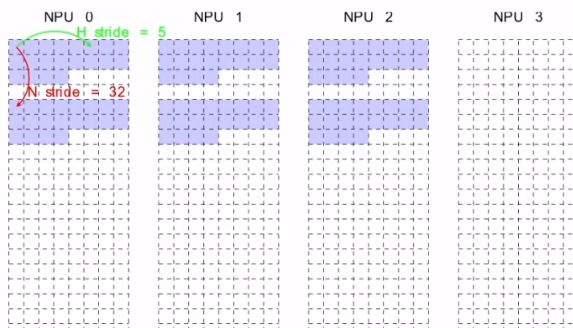
Remarks

- The data type of the destination and source tensors is 32-bit.
- If `dst_stride` is NULL, the destination tensor is in the Continuous Layout.
- If `src_stride` is NULL, the source tensor is in the 128-Byte Aligned Layout.
- `dst_stride->w` and `src_stride->w` are only allowed to be one.

内存分布

SOP

对于int8 可以有4n模式



N = 2, C = 3, H = 4, W = 5, fp32

- 每个 block 放一个 fp32
- C 维度上把数据切分到各个 NPU
- 注意 N stride 需要跨后面的 NPU

W stride 1
H stride 5
C stride 32
N stride 32

常用函数

```
void okk_initialize()
void okk_poll() 保证之前的错做完程
void okk_parallel_start() 控制并发
void okk_parallel_end() 这两个之前的操作并发执行，之外的就是串行
bool okk_is_parallel_state() 并发状态

unsigned int okk_local_mem_size_per_npu() 内存大小
unsigned int okk_l2_sram_size()
unsigned int okk_dtcn_size()

int okk_npu_num()
int okk_start_npu_index(local_addr_t addr)
int okk_channle_num_per_npu(int start_idx, int num_channels)
```

常用函数

给定数据尺寸和对齐要求，返回stride

SOP

```
void okk_128_byte_aligned_stride_for_32bit(
    dim4 *stride, int start_idx, constdim4 *shape)
void okk_128_byte_aligned_stride_for_16bit(
    dim4 *stride, int start_idx, constdim4 *shape)
void okk_128_byte_aligned_stride_for_8bit(
    dim4 *stride, int start_idx, constdim4 *shape)
void okk_compact_stride(dim4 *stride, int start_idx, constdim4 *shape)
void okk_continuous_stride(dim4 *stride, constdim4 *shape) 获取DDR或HOST中连续摆放
```

LOCAL Mem的stride

Cmodel 和 BMLib

pcie
soc 平级

CModel 是芯片所有功能在 x86 上用 c 语言的实现。

- 所有的原子操作都和硬件的实现效果一致
- 方便调试算子

BMLib 主要提供管理设备句柄以及 DDR 分配与释放的接口。

- Cmodel、pcie、soc 各个版本都不一样
- Bmkernel host 端编程的依赖

